Algorithmic Complexity & Theory

Big-O Time/Space Complexity, Sorting, Searching & Core Algorithms

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q1. 1. #include<iostream> using namespace std; class Test { static int x; public: Test() { x++; } static int getX() { return x; } }; int Test::x=0; int main() { cout<< Test::getX()<<" "; Test t[5]; cout<< Test::getX(); }

(A) 0 0
(B) 5 5
(C) 0 5
(D) a) Run time error

Correct Answer: Option (C) : 0 5

Logic / Derivation: Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (0 5).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q2. 1. #include<iostream> using namespace std; class Point { Point() { cout<<"Constructor called"; } }; int main(){ Point t1; return 0; }

(A) Constructor called
(B) Compile time error
(C) Run time error
(D) Blank line

Correct Answer: Option (B) : Compile time error

Logic / Derivation: Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (Compile time error).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q3. 1. #include<stdio.h> using namespace std; int main() { for (int x = 10; x >= 0; x--) { int z = x & (x >> 1); if (z) printf("%d ", x); } }

(A) 7 6 3
(B) 7 5 4
(C) 6 7 8
(D) 6 5 4

Correct Answer: Option (A) : 7 6 3

Logic / Derivation: Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (A) (7 6 3).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q4. 1. What will be the output of this statement? printf ("%d", 9/5);

(A) 1.8
(B) 1.0
(C) 2.0
(D) None of these

Correct Answer: Option (D) : None of these

Logic / Derivation: Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (D) (None of these).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q5. 1. What will be the output of this program? #include<stdio.h> int main() { int i = 5; printf("%d", i = ++i == 6); return 0; }

(A) 2
(B) 6
(C) 4
(D) 1

Correct Answer: Option (D) : 1

Logic / Derivation: Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (D) (1).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q6. 1. What will be the output of this program? #include<stdio.h> enum flg{a, b, c}; enum glf{c, e, f}; main() { enum flg h; h = b; printf("%d", h); return 0; }

(A) 1
(B) error: redeclaration of an enumerator
(C) h
(D) 3

Correct Answer: Option (B) : error: redeclaration of an enumerator

Logic / Derivation: Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (error: redeclaration of an enumerator).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q7. 1. What will be the output of this program? main () { if (5 < '5') printf("5") else printf(" NOT EQUAL TO 5.") }

(A) ENQ
(B) 5
(C) I
(D) Not equal to 5

Correct Answer: Option (B) : 5

Logic / Derivation: Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (5).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q8. 1. The value of j at the end of the execution of the following pseudo code is ? Integer incr(Integer i) Static Integer count = 0 count = count +i return count Main() Integer i,j for(start i from 0 to 4) j= incr(i) end for

(A) 10
(B) 4
(C) 6
(D) 7

Correct Answer: Option (A) : 10

Logic / Derivation: Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (A) (10).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q9. 1. The value returned by the following function for f(1) is? int f(int n) { static int i =1; if (n>=5) return n; n=n + i ; i++; return f(n); }

(A) 5
(B) 6
(C) 7
(D) 8

Correct Answer: Option (C) : 7

Logic / Derivation: Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (C) (7).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q10. Find the output of the following code. #include <stdio.h> int main(void) { int a = 10, x; x = a-- + +445; printf ("%d", x); return 0; }

(A) 19
(B) 435
(C) 455
(D) 23

Correct Answer: Option (C) : 455

Logic / Derivation: Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (C) (455).

[Campus Monk Day 06 Mock Test]
Topic: Campus Monk Mock Pseudocode
Q11. Find the output of the following code. #include <stdio.h> void main() { int a = 10; if (a == 010) printf("BCA"); else printf("B.Tech"); }
(A) BCA
(B) B.tech
(C) 10
(D) Error
Correct Answer: Option (B) : B.tech
Logic / Derivation: Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (B.tech).

[Campus Monk Day 06 Mock Test]
Topic: Campus Monk Mock Pseudocode
Q12. Find the output of the following code. #include <stdio.h> void main () { int x = 91; x >>= 2; printf ("%d\n", x); }
(A) 182
(B) 91
(C) 22
(D) 82
Correct Answer: Option (C) : 22
Logic / Derivation: Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (22).

[Campus Monk Day 07 Mock Test]
Topic: Campus Monk Mock Pseudocode
Q13. What is the output of the following C code? #include <stdio.h> int main() { int num = 987; int rem; while(num != 0) { rem = num % 4; num = num / 10; } printf("%d", rem); }
(A) 0
(B) 1
(C) 10
(D) 5
Correct Answer: Option (B) : 1
Logic / Derivation: Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (B) (1).

[Campus Monk Day 07 Mock Test]
Topic: Campus Monk Mock Pseudocode
Q14. What is the output of the following C code? #include <stdio.h> int main() { float x = 0.0; long int y = 10; printf("%d", sizeof(y) == sizeof(x + y)); return 0; }
(A) 3
(B) 1
(C) 5
(D) 2
Correct Answer: Option (B) : 1
Logic / Derivation: Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (1).

[Campus Monk Day 07 Mock Test]
Topic: Campus Monk Mock Pseudocode
Q15. What is the output of the following C code? int f(int x, int *py, int **ppz) { int y, z; **ppz += 1; z = **ppz; *py += 2; y = *py; x += 3; return x + y + z; } void main() { int c, *b, **a; c = 4; b = &c; a = &b; printf("%d ", f(c, b, a)); return 0; }
(A) 18
(B) 19
(C) 21
(D) 22
Correct Answer: Option (B) : 19
Logic / Derivation: Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (19).

[Campus Monk Day 07 Mock Test]
Topic: Campus Monk Mock Pseudocode
Q16. What is the output of the following C code? #include <stdio.h> int main() { int a = 12; void *ptr = (int *)&a; printf("%d", *ptr); getchar(); return 0; }
(A) 12
(B) Compiler Error
(C) Run Time Error
(D) 0
Correct Answer: Option (B) : Compiler Error
Logic / Derivation: Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (Compiler Error).

[Campus Monk Day 08 Mock Test]
Topic: Campus Monk Mock Pseudocode
Q17. Q2.) #include<iostream> using namespace std; int main() { int a = 3, b = 2, c = 5; int result = (a++ << 2) + (--b << 1) - (c-- >> 1) + (a & b); cout << result; return 0; }
(A) 12
(B) 14
(C) 10
(D) 16
Correct Answer: Option (A) : 12
Logic / Derivation: Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (A) (12).

[Campus Monk Day 09 Mock Test]
Topic: Campus Monk Mock Pseudocode
Q18. int main() { int x = 9, y = 2, z = 6; int a = x & y | z; printf("%d", a); return 0; }
(A) Error
(B) 2
(C) 3
(D) 6
Correct Answer: Option (D) : 6
Logic / Derivation: Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (D) (6).